```
#include <stdio.h>
#include <stdlib.h>
void simulate(int pages[], int n, int nf, int mode) {
    int frames[10] = {0}, time[10] = {0}, faults = 0, next = 0;
    for (int i = 0; i < nf; i++)
        frames[i] = -1;
    printf("\n--- %s Page Replacement ---\n", mode == 0 ? "FIFO" : mode == 1 ? "LRU" :
"Optimal");
    for (int i = 0; i < n; i++) {
        int hit = -1;
        for (int j = 0; j < nf; j++)
            if (frames[j] == pages[i])
                hit = j;
        if (hit != -1) {
            if (mode == 1)
            time[hit] = i;
        } else {
            int rep = -1;
            for (int j = 0; j < nf; j++) if
            (frames[j] == -1) {
                rep = j;
                    break;
                }
            if (rep == -1) {
                if (mode == 0) {
                    rep = next;
                    next = (next + 1) % nf;
            } else {
                    int extreme = (mode == 1) ? i : -1;
                    for (int j = 0; j < nf; j++) {
                        int look = -1;
                        if (mode == 1)
                        look = time[j];
                        else {
                            for (int k = i + 1; k < n; k++)
                                if (pages[k] == frames[j]) {
                                    look = k;
                                    break;
                                }
                            if (look == -1) {
```

```
                    rep = j;
                    break;
                    }
                  }
                    if ((mode == 1 && look < extreme) || (mode == 2 && look > extreme)) {
                        extreme = look;
                                rep = j;
                    }
                }
            }
        }
        frames[rep] = pages[i];
        if (mode == 1)
                    time[rep] = i;
        faults++;
    }
    printf("Page %d -> [", pages[i]);
    for (int j = 0; j < nf && frames[j] != -1; j++)
        printf("%d%s", frames[j], (j < nf - 1 && frames[j+1] != -1) ? " " : "");
    printf("]\n");
  }
  printf("Total Page Faults (%s): %d\n", mode == 0 ? "FIFO" : mode == 1 ? "LRU" : "Optimal",
faults);
}
int main() {
  int n, nf, *pages;
  printf("1WA24CS218\n");
  printf("Enter number of pages: ");
  scanf("%d", &n);
  pages = (int*)malloc(n * sizeof(int));
  printf("Enter page reference string:\n");
  for (int i = 0; i < n; i++)
        scanf("%d", &pages[i]);
  printf("Enter number of frames: ");
  scanf("%d", &nf);
  for (int m = 0; m < 3; m++)
        simulate(pages, n, nf, m);
  free(pages);
  return 0;
}
```

```
1WA24CS218
Enter number of pages: 15
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3 1 2 0
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]
Page 1 -> [0 1 3]
Page 2 -> [0 1 2]
Page 0 -> [0 1 2]
Total Page Faults (FIFO): 12
```

```
--- LRU Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]
Page 1 -> [0 3 1]
Page 2 -> [2 3 1]
Page 0 -> [2 0 1]
Total Page Faults (LRU): 12

--- Optimal Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [2 4 3]
Page 2 -> [2 4 3]
Page 3 -> [2 4 3]
Page 0 -> [2 0 3]
Page 3 -> [2 0 3]
Page 1 -> [2 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Total Page Faults (Optimal): 8
```